

Architectural Particular Specification

SUSPENDED CEILING SYSTEM

SECTION 8.6

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SECTION 8.6

SUSPENDED CEILING SYSTEM

PART 1 GENERAL

1.01 SCOPE OF WORK

- A. General: Provide ceilings in accordance with the Contract Documents.
- B. Section Includes: The Work of this Section includes, but is not limited to, the furnishing and installation of the following:
 - 1. Suspended metal grid ceiling system and adjustable hangers
 - 2. Ceiling panels
 - 3. Perimeter trims and custom trims/escutcheons for ceiling penetrations.
 - 4. External Ceiling System

1.02 REFERENCES

- A. British Standards :
 - 1. BS 8290 : Part 1: 1991 - Suspended Ceiling: Code of Practice for Design
 - 2. BS 8290: Part 2: 1991 - Suspended Ceiling: Specification for Performance of Components and Assemblies.
 - 3. BS 8290 : Part 3 : 1991 - Suspended Ceiling : Code of Practice for Installation and Maintenance
- B. American Standards :
 - 1. ASTM C635 - 92: Practice for Installation of Metal Ceiling Suspension Systems for Acoustical Tile and Lay-in Panels.
 - 2. ASTM C635 - 94: Specification for the Manufacture, Performance, and Testing of Metal Suspension Systems for Acoustical Tile and Lay-in Panel Ceilings.

1.03 QUALITY ASSURANCE

- A. The ceiling work shall be performed by a firm having 5 years experience in the installation of materials specified herein on projects comparable to this Project. The firm shall have the approval of the materials manufacturer.
- B. Single-source Responsibility: Obtain each type of ceiling system from a single source with resources to provide products of consistent quality in appearance and physical properties without delaying progress of the Work.
- C. Qualification of Manufacturers: Company specializing in the manufacturer of commercial grade panel ceiling and suspension systems with minimum of ten years documented experience.

1.04 PERFORMANCE REQUIREMENTS

- A. All ceilings, whether horizontal, vertical or inclined, shall be rigid enough to provide true and flat surfaces under the calculated ceiling loads, including loadings from all mechanical electrical and fire-fighting fixtures and to be able to withstand lateral movement. No deflection under the maximum loadings shall exceed 1/360 of the span.
- B. The Contractor shall make provision for all additional hangers for air-conditioning grilles, fire-fighting fixtures, ceiling exhausts, light troughs and fittings and the like.
- C. The Contractor shall make provision for all additional hangers for air-conditioning grilles, fire-fighting fixtures, ceiling exhausts, light troughs and fittings and the like.
- D. The Contractor shall provide all materials and fittings and perform any work which is obviously necessary for the complete ceiling installation even though such materials, fittings or works may not be explicitly mentioned in this specification or shown in the drawings. The supply of all components for the final installation and handing over in good condition of the works shall be the sole responsibility of the Contractor.
- E. The Contractor shall check all actual dimensions of the ceiling systems based on final layout plans and other working details.
- F. The ceiling systems shall satisfy the performance requirements and include, inter alia, the following : -
 - 1. Reflected ceiling plans showing the configuration and layout of all ceilings;
 - 2. Layout and design of suspension members;
 - 3. Infill and access panels;
 - 4. Assembly and all construction details;
 - 5. Junction details between ceilings and walls;
 - 6. Trimming details around openings;
 - 7. Construction details at changes in level
- G. The Contractor shall also be responsible for : -
 - 1. Schedule and monitoring of the works,
 - 2. Co-ordination with works of other trade,
 - 3. Hoisting and temporary site work.
- H. The Contractor shall furnish all labour and material, equipment and services necessary for the reasonably incidental to the furnishing and complete installation of all suspended ceiling works as shown on the drawings and / or as specified herein.
- I. Installation of all suspended ceiling works shall be in strict accordance with the manufacturer's specifications.
- J. The finished works shall be smooth, clean and free from damage and defects.

- K. Unless otherwise specified, all materials shall be of the best quality of the types specified.
- L. Non Standard Modules and Edge Details
All design criteria is for standard modules. Refer to drawings for variations from standard in size, shape, profile and aperture.
- M. Total Flexibility
Generally all ceiling panels shall be removable to allow access to the ceiling void above.
- N. Wall Angle
Wall angles shall be of electrolytically galvanised fine sheet steel as shown on drawings and finished to match the ceiling finishes.
- O. Colour Consistency
Colour consistency should be accurate, especially within the same floor, and the Contractor shall make every effort to achieve uniformity of colour. Components with colour deviations will be rejected.
- P. Levelling and Setting Out

The Contractor shall be required to set out the top and bottom levels of the suspended ceiling at sufficient locations for reference by other trades.

The Contractor shall also be required to set out the basic ceiling grids at site with fishing lines and sufficient reference marks for reference by other trades for the installation of lighting fixtures, sprinkler heads, A/C grilles, smoke detector, etc.
- Q. Tolerance
All work shall be plumb, square, level, and correctly aligned. The acceptable tolerance shall be to BS 5606:1978
- R. Measurement verification
The Contractor shall check all dimensions on site and be responsible for his own inaccuracies of measurement.
- S. Co-ordination with other trades
Work specified in other sections and requiring coordination by the Contractor includes, but is not limited to, the following:-
1. Wall tiling
 2. R.C./block wall structure
 3. Window & glass partition wall system
 4. Sprinklers, heat/smoke detectors, light fittings, P.A. speakers, air diffusers and return air grilles, thermostats and other M/E building services
 5. Fire shutters and security shutters
- T. The Contractor is required to carry out the coordination and the setting out of all services and other items which are to be fixed and connected to or directly adjacent to the Suspended Ceiling System so that all service installations and work of other trades over grids shall be completed prior to fixing of finished ceiling materials.

- U. The Contractor shall acquaint himself with the layout of services contained in the ceiling void, and is deemed to have full knowledge of the layout of all services such as air conditioning ducts and sprinkler pipes and to have allowed for all necessary adaptation to the suspension system in order to span and clear such services.
- V. The finished ceiling is required to accommodate sprinkler heads, heat detectors, P.A. outlets, thermostats, diffusers, downlights, speaker, CCTV etc.
- W. Air diffusers & return air grilles shall be supplied by A/C Specialist Contractor. The Contractor shall coordinate to ensure that the colour coating of the A/C fittings matches to the highest degree with the colour and finish of the metal panels.
- X. Ceiling panels shall be factory trimmed ceiling panels to accommodate these. The Contractor shall be required to liaise closely and coordinate the installation work of each sprinkler head, heat/smoke detector, thermostat, diffuser, P.A. speaker and special downlight, with the ceiling system in order that no site trimming is needed.
- Y. Site cutting and trimming is not allowed except with the written approval of the Architect. All ceiling panels shall be designed and factory fabricated to accommodate service cut outs.
- Z. Unless otherwise specified, the contractor can submit system or product conforming to the above specified performances to the Architect for approval. No claim for any additional cost or time for the use of any alternative as proposed by the contractor will be entertained by the Architect.
- AA. Any cost saving as a result of the Architect's acceptance for any alternative system or product shall be adjusted according to the contract conditions.
- BB. Due allowance shall be made by the contractor to ensure all necessary ordering and delivery works will be carried out in accordance with the contractor's master programme.
- CC. Construct suspended ceiling in accordance with British Standard (BS) 8290: Part 2. Fix proprietary systems in accordance with the manufacture's recommendations. Ensure that ceiling tiles and grids are properly set out and that all cutting is done at the perimeter. Fix all cover strips, edge trims and the like.
- DD. Verify all as-built dimensions and conditions on site before commencing the ceiling works. The highest workmanship is required and all works shall be carried out by skilled labour in an efficient and expedient manner.
- EE. Ceilings shall be constructed square and true to plane, both in themselves and in relationship with each other where installed in a continuous plane. All components and materials shall be carefully handled at work and on site, both during delivery or handling.

1.05 ENVIRONMENTAL REQUIREMENTS

Do not install materials unless temperature and humidity conditions closely approximate the interior conditions which will exist when the building is occupied. Maintain temperature and humidity conditions before, during, and after installation. Plastering and concrete work (including grinding) shall be complete and dry. Windows and door shall be in place and glazed.

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

- A. The Contractor and Manufacturer shall be aware of the Indoor psychometric parameters used by the Consulting Mechanical Engineer in determining vapour barrier, insulation, indoor temperature and relative humidity requirements, heating and cooling loads.
- B. The Contractor and Ceiling System Manufacturer shall warrant that all products, materials and items are suitable, and can be maintained in the various climatic and indoor psychometric conditions encountered without acoustical ceiling panel degradation; surface condensations of ceiling, and support and growth of moulds.

1.07 SUBMITTALS

- A. Shop Drawings:
 - 1. Submit details and reflected ceiling plans of ceilings before proceeding with Work. Provide coordination drawings for reflected ceiling plans drawn accurately to large scale and coordinating penetrations and ceiling-mounted items. Show the following:
 - a. The ceiling layout relation to adjoining structure, mechanical and electrical system, and other related works,
 - b. Reflected plan of the ceiling showing ceiling grid and arrangement of components, positions of lights, diffusers, hatches and service penetrations.
 - c. Sequence and details of fabrication, assembly and erection, methods of achieving demountability.
 - d. Full scale details,
 - e. Dimension and thickness,
 - f. Specification of materials and finishes to be used, finishes to exposed members, corrosion protection, performance data of components and assemblies and any other pertinent information.
 - g. Location and spacing of screw, hangers, anchorage and other accessories, together with structural calculations if necessary.
 - h. All other pertinent information :
 - 1) Allocation of all M/E services, access panels on false ceiling.
 - 2) Large scale details of construction, suspension system, method of assembly, trim and fixing, showing dimensions, clearances and tolerances.
 - 3) Method of attaching heads of partitions to the ceiling support members.

- 4) Method of reducing contact vibrations between structure and ceiling.
2. The shop drawings shall indicate the final ceiling levels after coordination of E&M services above the suspended ceiling. It is the contractor responsibility to try to achieve the Architect's intended ceiling levels. If the Architect's ceiling levels are required to be altered, the extent of the altered part shall be highlighted in the shop drawings with full justification for Architect's comment. Any subsequent amendments arise due to the alteration of ceiling level e.g. adjustment of Architect's details, additional bulkhead details or additional upward sprinkler heads shall be submitted together with the proposed alteration for Architect's comment.

The Architect will review the shop drawings to ensure their conformity to the contract documents. Architectural review of shop drawings shall not relieve the Contractor from any of the responsibilities and requirements as herein specified.

The Contractor shall lay out work in accordance with the drawings and check drawings of other trades to verify spaces in which work will be installed. Maximum headroom and space condition shall be maintained at all points.

If directed by the Architect, the Contractor shall without extra charge, make reasonable modifications in the layout as needed to prevent conflict with work of other trades or for proper execution of work.

B. Samples:

1. The Contractor shall submit samples and performance data of materials and articles to be used in the work for Architect's comment.
2. The order for bulk supplies of materials and articles shall not be confirmed until confirmation of no adverse comment from the Architect.
3. The sample shall at least 1.2m x 1.2m module units of the ceiling system, including all related accessories such as ceiling grid, hangers, trims etc.
 - a. Ceiling Material: Sheet, panel, tile, strip, with insulation where applicable, showing the extremes and mean of any variation in colour, pattern or texture of the proposed finish.
 - b. Suspension: Sections proposed to be used for suspension system, including wall angles and trim, finished as proposed.
 - c. Accessories: Visible accessories required as part of each ceiling installation, such as light fittings, diffusers, detectors, hatches and curtain tracks.

- C. The Contractor is responsible for making all enquiries necessary to confirm the acceptability of materials used, and the installation, from the Fire Services Department. Where materials are required to be flameproof the requirement shall be deemed to have been included in the Contract figure.

- D. Product Data: Submit manufacturer's specifications and installation instructions for each acoustical material, suspension system and other products required, including certified laboratory test reports and other data as may be required to show compliance with the Contract Documents.

1.08 DELIVERY STORAGE AND HANDLING

- A. Deliver materials to the Project site in manufacturer's unopened containers, clearly indicating manufacturer's name, brand, type, style, size, color, texture and other identifying information.
- B. Store materials in a dry location, off the ground and in a manner to prevent damage, deterioration and intrusion of foreign matter. Replace materials which have been damaged or are otherwise unfit for use.
- C. Before installing acoustical ceiling units, permit them to reach room temperature and stabilized moisture content.

1.09 MOCK-UP

- A. This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.10 PRE-INSTALLATION CONFERENCE

- A. Review foreseeable methods and procedures related Suspended Ceiling Work including the following:
 - 1. Tour, inspect and discuss condition of substrate, anchorage and interface with other finish/material, penetrations and other preparatory work performed by other building trades.
 - 2. Review requirement for Suspended Ceiling System works (Drawing, Specifications and other Contract Documents).
 - 3. Review required submittals, both completed and yet to be completed.
 - 4. Review and finalize Construction Programme of Work related to this Work and verify availability of materials, installer's personnel, equipment and facilities needed to make progress and avoid delays.
 - 5. Review required inspections, testing, certifying and material usage accounting procedures.
 - 6. Review weather and forecasted weather conditions, and procedures for coping with unfavourable conditions, including possibility of temporary protections.
 - 7. Coordinate layout and installation of ceiling systems and metal pan ceiling systems with other construction that penetrates ceilings or its supported by them, including light fixtures, HVAC equipment, fire-suppression system, and partition assemblies.

1.11 WARRANTY

- A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.
- B. Warranty specified in this Article shall not deprive the Employer of other rights the Employer may have under other provisions of the Contract Documents and are in addition to and run concurrent with other warranties made by the Contractor under requirements of the Contract Documents.

1. The Contractor, Panel Ceiling Manufacturer and Suspension System Manufacturer shall jointly provide a five (5) year warranty against material defects and workmanship for Work of this Section.
2. Warranty: Provide a written warranty, directly to the Owner, for a period of five (5) years warranting against delamination, discoloration, suspension system or gird failure, warping or sagging (exceeding 2.5mm at the centre point of the tile) resulting from defects of materials or factory workmanship and against the occurrence of 50% red dust. Upon notification of such defects, within the warranty period, make the necessary repairs and replacements, at the convenience of the Owner.

1.12 SPARE PARTS

- A. This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.
- B. Furnish to the Employer and store all the site of each type of each colour and each model or product installed in the Project, packaged in manufacturer's unopened cartons and identified as to contents.
- C. Provide 4 no. sealed containers of touch-up paint for ceiling grid and metal work minor scratches.

PART 2 PRODUCTS

2.01 MATERIALS

To provide the following ceiling system and all associated accessories as shown in relevant Contract Drawings and Schedules:

A. Unistruct Supporting Ceiling Frame System

1. Proprietary “Unistruct” supporting frame system @ 1200mm x600mm 4-ways grid, and, hanging from the Secondary Beams at the Ceiling of the New Computer Rooms shall be supplied and installed.
2. The whole “UNISTRUCT” supporting frame system shall be able to provide a Service Load of 2.4 kPa.
3. All necessary fixing accessories shall be allowed and incorporated in the system for holding and fixing the specified metal panel ceiling system between the 1200mm “UNISTRUCT” channels grid.
4. The detail arrangement and required fixing components for the system shall be proposed by the Contractor to the satisfaction of the Architect.
5. Structural calculations to demonstrate the structural integrity of the whole supporting frame system shall be prepared by an independent professional engineer and submitted for approval by the Project Structural Engineer.
6. The system shall be provided with 1200x600mm suspended Unistruct ceiling supporting frame with aluminum infill panels and air grille.

B. Fire Resisting False Ceiling and Enclosure

1. General
 - a. All fire resisting false ceiling and enclosure ceiling/enclosure shall be made of fire resisting panel or equivalent and shall satisfy the requirement of BS476: Part 8 and 22 for the specified fire resistance period.
 - b. All construction joints shall not be greater than 3mm unless these joints are sealed up by fire resisting sealant approved by Architect.
 - c. All construction methods and details shall strictly follow the requirement/specification of the manufacturer.
2. Fire Resisting Single Frame Metal Stud Ceiling System
 - a. Fire resistance:
2-hours FRP, Integrity and Insulation, E&M services enclosure in accordance with BS476 Part 20: 1987
 - b. Board: 1220 x 2440 x 9mm fire resisting fibre cement board
 - c. Construction:
 - 1) KNAUF Superpanel 9mm thick or approved equivalent fire resisting fibre cement board
 - 2) Hangers fixed by push-in anchor to limit stress $\leq 10\text{N/mm}^2$, Max 1500mm spacing
 - 3) 50x50x0.6mm thick. Steel angle at corners & wall & slab

- 4) M4 self-tapping screw at nominal 200mm c/c
 - 5) Steel channel collar min 32mmx50mmx0.6mm at butt joints of nominal 1220mm centres
 - 6) Fire resisting fibre cement board fillet-9mm thick x 75mm(w) fillet fixed by screw (inside/outside)
 - 7) E&M services to be independently supported
 - 8) M6 anchor bolt fixed to wall and slab at 500mm c/c
 - 9) 50mm to 70mm thick x 100kg/m³ Rockwool
 - 10) Steel channel 32mmx50mmx0.6mm
 - 11) 1mm thk. Steel channel blacket fixed by 2nos. M4 screw
 - 12) Fire resisting fibre cement board 75x75x9mm thk.cover plate & Fire Sealant
 - 13) 1mm thick.Steel Angle blacket fixed by screw to concrete slab by M8 anchor bolt
 - 14) Fire Sealant.
- d. Approved Manufacturer / Supplier:
- KNAUF Hong Kong Limited or approved equivalent
 Thomas, K.H. Cheung
 Tel: 29649311 Mobile: 91878301

C. Gypsum Board Ceiling System

1. Board Lining

a. Standard Gypsum Plasterboard

- 1) Description: Knauf Regular Gypsum Board or approved equivalent
- 2) Size: 1220 mm x 2440mm x 12.5mm thick
- 3) Density: 725kg/m³
- 4) Asbestos content: None
- 5) Appearance: Grey Paper Liner
- 6) Complied standard:
 EN 520 – Type A
 BS 1230 – Type 1
 BS 476 Part 4, 6, 7
 EN 13501 – 1
 ASTM C1396
 ISO 9001

b. Water Resistant Gypsum Plasterboard

- 1) Description: Knauf Moistureshield Gypsum Board or approved equivalent.
- 2) Size: 1220 mm x 2440mm x 12mm thick
- 3) Density: 730kg/m³
- 4) Asbestos content: None
- 5) Appearance: Green Paper Liner

- 6) Surface Water Absorption: $EN520 \leq 180g/m^2$
- 7) Total Water Absorption: $EN520 \leq 5\%$
- 8) Complied standard:
 EN 520 – Type DF
 BS 1230 – Type 5
 BS 476 Part 4, 6, 7
 EN 13501 – 1
 ASTM C1396
 ISO 9001

2. Framework to Ceilings

- a. For light duty ceiling with 1 layer of 12.5mm thick KNAUF Regular plasterboard, KNAUF LD-19 ceiling suspension system or approved equivalent, shall be used.
- b. According to the weight of the ceiling (kg/m^2) inclusive of lighting / ventilation system fitting, with clips, brackets perimeter channels, and other proprietary accessories as manufactured by KNAUF, or acceptable equivalent, shall be used.
- c. The ceiling board joints shall be filled with KNAUF Ready Mix Joint Cement and taped with KNAUF paper joint tape in order to create a flush and seamless surface, or approved equivalent. Both the joint cement and tape shall be of the same brand and manufacturer as the applied gypsum board ceiling system.

3. Approved Manufacturer / Supplier:

KNAUF Hong Kong Limited or approved equivalent
 Thomas, K.H. Cheung
 Tel: 29649311 Mobile: 91878301

D. Mineral Fiber Ceiling Tile System

The ceiling system shall include mineral fiber ceiling tiles, aluminum suspension systems and accessories, including but not limited to moldings, perimeter trim, clips and edge caps, all from a single manufacturer.

1. Mineral Fiber Ceiling Panels

- a. Brand: Armstrong (America) or approved equivalent
- b. Description: Ultima RH99 (Square Lay-in, Humidity Resistance 99%) Ceiling in White Color (Model: BP1001) or approved equivalent
- c. Panel size: 600mm x 600mm x 19mm thick
- d. The ceiling panel shall be made of wet-form mineral fiber with transparent membrane surface finish with factory-applied acylic latex surface finish.
- e. The ceiling system should achieve ISO Class 5 and ultra-low emissions VOC A+.
- f. The ceiling system shall be complied BS476 Part 6 & 7 Class 0 / Class 1, US Standard Class A – Flame Spread 25 or under.
- g. PVC hold down clip should be applied for lay-in board to ensure air tight ceiling property.

- h. Acoustical performance: The panels shall be achieved NRC not less than 0.7 and CAC 39 acoustic performance.
 - i. Humidity Exposure: The ceiling shall achieve humidity resistance RH99 and 38% recycling content
2. Exposed Tee Grid System
 - a. Brand: Armstrong (America) or approved equivalent
 - b. Description: PeakForm 888 24mm Exposed Tee Grid System for ANF RH90 Ceiling or approved equivalent
 - c. Size: Main Tee – 3000mm x 24mm x 35mm, Cross Tee – 1200mm x 24mm x 27mm, Cross Tee – 600mm x 24mm x 27mm
 - d. Material: Double-web hot dipped galvanized steel with baked polyester paint surface finish in white
 3. Approved Supplier:

Ngai Shun Decoration And Materials Co., Ltd.
 Unit C, 21F., Seabright Plaza,
 9-23 Shell Street, North Point, Hong Kong
 Contact Person: Allen Liu
 Mobile: 9173 4949
 Tel: 2578 6038, 2571 4638, 2571 8489
 Fax: 2802 6366
 Email Address: allen@nsdm.com.hk

E. Metalworks Mesh Ceiling System

1. Mesh Panel
 - a. Brand: Armstrong (America) or approved equivalent
 - b. Description: METALWORKS Mesh Scallops / Lattice Pattern Square Lay-in Ceiling System
 - c. Model No.: MEM1012/1012AR
 - d. Size: 600mm x 1800mm x 1.2mmT
 - e. Material: Cold-rolled Steel
 - f. Surface Finish: Powder Coated finish
 - g. Fire Performance: Class A: ASTM E84 and CAN/ULC S102; Flame Spread Index 25 or less. Smoke Developed Index 50 or less. All Metalworks Mesh Panels Are Non-combustible.
 - h. Color: White
2. Exposed Tee Grid System
 - a. Brand: Armstrong (America) or approved equivalent
 - b. Description: Prelude XL 360° Painted Grid or approved equivalent
 - c. Material: Hot dipped galvanized steel
 - d. Surface Finish:
 - 1) 2.0 mil powder coated painted surface or baked polyester paint
 - 2) Manufactured and tested in accordance with ASTM C635
 - e. Cross Tee/Main Beam Interface: Override

- f. End Detail:
 - 1) Main Beam: Staked-on clip
 - 2) XL® Cross Tee: Staked-on clip
 3. Approved Supplier:
 Ngai Shun Decoration And Materials Co., Ltd.
 Unit C, 21F., Seabright Plaza,
 9-23 Shell Street, North Point, Hong Kong
 Contact Person: Allen Liu
 Mobile: 9173 4949
 Tel: 2578 6038, 2571 4638, 2571 8489
 Fax: 2802 6366
 Email Address: allen@nsdm.com.hk
- F. Non-perforated Aluminium Suspended Ceiling System
1. Brand: Dampa 300 or approved equivalent
 2. Provide materials complying with the following performance requirements, unless otherwise specified:
 - a. Light Reflectance: 15-20% gloss level.
 - b. Fire Propagation / Spread of Flame : BS476:Part 6, Part 4
 - c. Non-Combustible: BS476-part 4, BS EN 13501
 3. Metal Panel
 - a. Non-Perforated Dampa 300, Size 300mm width, Carrier #40

 Aluminium Stripe Panel Ceiling System
 Brand: Chicago Metallic
 Series: Dampa 300 Square Edge Ceiling
 Location: Lavatories, Shower Room, Pantry or other as specified in drawings.
 - b. Size: 300mm wide
 - c. Suspension System: Conceal Suspension System
 - d. Finish: Pre-painted Aluminium Coils in Standard White Colour
 - e. Appearance: Plain
 - f. Related Fire Certificates & Reports are required to submit & approve by Architect.
 - g. Description:
 The System shall consist of roll-formed perforated strip panel clipped onto proprietary DAMPA® modular carrier No. 40. Panels shall be formed with upward bending at both sides and further inward bending flange on one side and the opposite outward bending flange. All edges at four sides shall be 'square'. Ceiling panels shall be in modular distances of 300m. (nominal), with 20mm groove. Length shall be tailored made to suit layout.
 - h. Perforation Pattern

 All aluminum panels shall be unperforated.

- i. Metal of Ceiling
Panels shall be manufactured from 0.7mm. thick aluminum.
 - j. Production
 - Aluminum coil shall be decoiled, levelled, baked enamel painted and roll-formed to finished panels.
 - Panels shall be manufactured in accordance with TAIM regulations.
 - k. Surface Finish
 - 1) Pre-painted aluminum coil in PE coating shall be decoiled, levelled, and roll-formed to finished panels.
 - 2) Panels shall be factory finished with paint, polyester epoxy in matt finish with an average 15-20% gloss level. The thickness of the coating shall be approx. 20-25 microns.
 - 3) Standard matt surfaces can produce a total light reflection level of up to 86 per cent depending on the colour.
 - l. Colour
Standard white colour CMC-03 White Colour.
 - m. Quality Standard
All "Chicago Metallic – DAMPA" Metal Ceilings are manufactured according to ISO 9001:2008 & ISO 14001:2004.
4. Primary Suspension Members
- a. Provide ceiling support components complying with ASTM C635 Classification "Light Duty", unless otherwise specified.
 - 1) Carrier System Grid #40 for ceiling panel type: Dampa 300 with perimeter wall angle trim to manufacturer's details, including:
 - (i) Accessories: Stabilizer bars, clips, and splices connector recommended by Ceiling Manufacturer.
 - (ii) Grid Finish Matte: Black Colour.
 - (iii) Support System: cold formed steel channels; galvanized adjustable hanger channel straps.
 - (iv) Edge Trims: Metal channel with 15mm recessed groove. PPGI White Colour CMC-03, corrosion-resistant coating of 20 microns.
 - b. light fixtures, diffusers and other items occurring in or on the ceiling shall be suspended independently
 - c. Finish to lines and levels shown, with maximum deflection not to exceed 1/360 of the span between supports.
5. Metal Wall Moldings: Electro-galvanized roll formed steel with manufacturer's standard beaked-on white enamel coating to match finish of material and that fit type of edge detail and suspension system indicated.
- a. For lay-in panels with reveal edge details, provide stepped edge molding that forms reveal of same depth and width as that formed

between edge of panel and flange at exposed suspension member.

- b. For circular penetrations of ceiling, provide edge moldings fabricated to diameter required to fit penetration exactly.
 - c. For narrow faced suspension systems, provide suspension system manufacturer's standard edge moldings that match width and configuration of exposed runners.
6. Tinline Exposed Mechanical Suspension System Ceiling: Manufacturer's standard design; complying with ASTM C635; complete with hangers, main tees, cross tees, splices, angle molding, and accessories. Provide components as follows:-
- a. Primary Suspension Members
 - b. Main Tees: 1-1/2 in. high by 9/16 in. wide formed of 0.025 in. galvanized cold-rolled steel section and bottom flange, faced with a roll formed galvanized steel cap.
 - c. Metal Wall Trimming, stepped where tegular edged tile is specified.
 - d. Accessories: Galvanized steel, specifically designed for use with the main components.
 - e. Exposed Finish: Chemically clean, electro-galvanize and bonderize all rolled formed parts. Finish with a white baked-on enamel coating, to match finish of metal ceiling unless otherwise shown.

PART 3 EXECUTION

3.01 EXMANINATION / INSPECTION

- A. Examine substrates and structural framing to which ceiling system attaches or abuts, with Installer present, for compliance with requirements specified in this and other sections that affect installation and anchorage of ceiling system. Do not proceed with installation until unsatisfactory conditions have been corrected.
- B. Coordination: Furnish layouts for preset inserts, clips, and other ceiling anchors whose installation is specified in other sections. Furnish concrete inserts and similar devices to other trades for installation well in advance of time needed for coordination of other works.

3.02 PREPARATION

- A. Work area to be examined before beginning work to determine that it is properly enclosed and the structure is in proper condition to receive the suspension system. Area shall be broom cleaned and uninterrupted for free movement of rolling scaffolds. Work shall not proceed until recommended job conditions have been assured.
- B. The Contractor is to be responsible for all necessary mild steel framing hangers, trapeze assemblies and/or supports in cases where the suspension system is obstructed by other trades and cannot be fixed directly to the structural soffit.
- C. The Contractor shall be responsible for handling and relocating all his materials and equipment required for the completion of the work stored on site to appropriate areas in order not to obstruct other trades.
- D. The Contractor shall also protect existing work from damage during the installation and make good any damage.

3.03 INSTALLATION

- A. General Installation
 - 1. All finish ceilings shall be laid true to line and level, with accurately aligned joints, set to proper finished ceiling height and rigidly fastened. No waves, bends, off sets, or irregularities shall be permitted.
 - 2. If, in the opinion of the Architect, any ceiling installation has not been made in strict accordance with these Specifications and the approved shop drawings either before or after final erection, the Architect may require the removal of the installation and require its replacement with new material.
 - 3. All ceilings shall be left fair-faced, free from all marks, grease, scratches, finger-marks or other blemishes and shall be handed over in a clean and perfect condition to the satisfaction of the Architect.
 - 4. All workmanship shall be first class in every respect. Only skilled workmen, experienced in this trade, under competent supervision shall be employed in the installation of the suspended ceilings. All work shall be carried out in accordance with the applicable portions of B.S. CP290 and to the satisfaction of the Architect. The Contractor shall

form accurately and without damage all holes required for light fixtures. Additional carrier, channels, clips or special support pieces shall be fixed as required for lighting fixtures, ventilation and air conditioning grilles, or other such services incorporated in the ceiling. Such work shall be done in cooperation with these other trades.

5. The carriers and other such support pieces shall be fixed to the hangers and shall be set true and level to proper heights above the floor allowing for finish, and shall preclude any bowing or sagging. Maximum allowance deflection of any support member shall be $1/360$ of the span, unless otherwise directed by the Architect.
6. Suspended grids must be inspected and approved by the Architect before commencing installation of finish ceiling materials.
7. Following installation, the Contractor is responsible to clean soiled or discoloured surfaces of units, and to remove and replace units which are damaged or improperly installed. The cost of replacement shall be deemed to be contractors' responsibilities, and no claim to the Employer for extra cost will be entertained.
8. Grid Members:
Where required, notch grid members at the junction with the perimeter trim to ensure the panels lie flat on the perimeter trim.
9. Fastenings:
Provide fastenings which are not visible in the finished ceiling.
10. Bulkheads:
Construct bulkheads and other similar ceiling formations to the profiles shown on the drawings so that they are an integral part of the ceiling structure and comply with the ceiling performance requirements.
11. Support Members:
Space the support members as required by the loads on the system and the type of ceiling and allow for the installation of services and accessories, including ductwork, light fittings and diffusers. Provide additional back support or suspension members necessary for fixing of such items. Do not suspend form services (e.g. ductwork) unless the service has been designed to accept the ceiling load. In locations where services obstruct the ceiling supports, provide bridging and suspension on each side of the services.
12. Bracing:
Provide bracing where necessary to prevent lateral movement.
13. Plasterboard:
Plasterboard Ceilings shall be constructed with 2 layers of 12mm thick gypsum board, staggered at joints. Fit panels accurately and neatly, free from air leakage and staining. Provide additional support or bracing to panels which are required to carry dead loads other than the panel's own weight.
14. Ceiling Grid:
Set out the ceiling grid as set out on the drawings and so that panel joints and centrelines of visible suspension members coincide with grid lines shown. If not otherwise shown, set out so that opposite margins are equal.
15. Pattern And Texture:
Set out patterned or heavily textured materials to give consistency in direction of pattern or texture.

16. Approval:
Obtain approval of the set out from the Architect before commencing the installation.
17. Special Sized Panels:
Provide special sized panels where necessary to fill non-standard margins, openings, penetrations and the like, purpose made unless cut panels are approved. Generally only plain units or units within a random pattern will be considered for approval to cut.
18. Provide access panels in non-demountable ceilings where shown on the drawings, supported and anchored by methods which permit ready removal and refixing.
19. Finish:
Provide access panels which match the ceiling panels in appearance and performance.
20. Reinforcement:
Reinforce the back of the access panels by an approved means to prevent warping and facilitate handling.
21. Provide movement joints in sheet finishes where shown on the drawings or at centres as recommended by the ceiling lining manufacturer. Where possible position joints to intersect lighting fixtures, vents or air diffusers.
22. Provide movement joints with purpose made control joint beads.
23. Provide trim at junctions with other building elements and surfaces, such as walls, beams and penetrations, consistent with the style, materials and finishes of the ceiling system generally.
24. The Contractor shall correct any installation defects as directed and within the time frame directed by the Architect and leave the installation clean and fully functional.

B. Installation of Mechanical Suspension System

1. Install primary suspension members and mechanical suspension system in accordance with ASTM C636 and manufacturer's written instructions to support required loads and to prevent deflection in excess of 1/360 of the span between supports. Water or laser level accurately in both directions, leveling to a tolerance of 3mm in 3600mm.
2. Install suspension systems with hangers supported only from building structural members. Locate hangers not less than 150mm from each end and spaced 1200mm maximum along each carrying channel or runner. Coordinate spacing of hangers, carrying channels, runners and molding with the location of electrical fixtures, sprinkler heads, and other items occurring in or on the ceiling. Splay hangers only where required to miss obstructions and offset horizontal force by bracing or other approved methods.
3. Where ceilings are suspended below ductwork, piping or other building elements which are not suitable for ceiling attachment due to strength limitations, restrictions of local authorities having jurisdiction, or ceiling system manufacturer's limitations, provide additional supplemental framing, supports and related work as required to span beneath these elements from suitable support locations. Design supplemental framing and supports to accommodate the spans and loads to be

sustained and to limit maximum deflections to the criteria specified and to finish to the lines and levels shown.

C. Installation of Metal Ceiling

1. Install ceiling panel clipping into the main runner. Cut and fit ceiling panel neatly against abutting surfaces and penetrations. Support edges by wall trimming.
2. Install hold-down clips in areas shown and in areas where required by government regulations or for fire-resistance ratings; spacing as recommended by panel manufacturer.

3.04 CLEANING

- A. Clean and repair surfaces that have been stained, marred, or otherwise damaged. Comply with manufacturer's instructions for cleaning and touch-up of minor finish damage. Remove and replace work that cannot be successfully cleaned and repaired to permanently eliminate evidence of damage.
- B. Final Clean Down - On completion of all the works, the Contractor is to carry out a complete final clean down of the ceilings. This final clean down is to be carried out in addition to any other cleaning that the Contractor may have done on a floor by floor or section by section basis.
1. Following installation, clean all soiled surfaces of units. This procedure is to be repeated on completion of installation & prior to final handover of the works by the Contractor.
 2. Remove and replace units which are damaged, discolored or improperly installed, to the satisfaction of the Architect.
 3. Dust and loose dirt shall be removed with a soft brush or vacuum cleaner. Cleaning must be in one direction only. Pencil marks, smudges or blemishes may be removed with an art gum eraser or with a moist cloth or sponge with mild soap, wiped off with a clean damp cloth or sponge.

3.05 PROTECTION

- A. Protect metal ceilings during the construction period so that they will be without any indication of deterioration or damage at the time of acceptance by the Owner.
- B. All damaged metal ceiling, grids edge trims and accessories shall be replaced at the contractor cost.

END OF SECTION